

using algorithms. The buy-side may specify which broker algorithms to use to trade single or basket orders, or rely on the expertise of sell-side brokers to select the proper algorithms and algorithmic parameters. It is important for the sell-side to precisely communicate to the buy-side expectations regarding expected transaction costs (usually via pre-trade analysis) and potential issues that may arise during trading. The buy-side will need to ensure these implementation goals are consistent with the fund's investment objectives. Furthermore, it is crucial for the buy-side to determine future implementation decisions (usually via post-trade analysis) to continuously evaluate broker performance and algorithms under various scenarios.

Quantitative, statistical arbitrage traders, sophisticated hedge funds, and the newly emerged class of investors known as high frequency traders will also program buying/selling rules directly into the trading algorithm. The program rules allows algorithms to determine instruments and how they should be bought and sold. These types of algorithms are referred to as “blackbox” or “profit and loss” algorithms.

For years, financial research has focused on the investment side of a business. Funds have invested copious dollars and research hours on the quest for superior investment opportunities and risk management techniques, with very little research on the implementation side. However, over the last decade, much of this initiative has shifted towards capturing hidden value during implementation. Treynor (1981), Perold (1988), Berkowitz, Logue, and Noser (1988), Wagner (1990), and Edwards and Wagner (1993) were among the first to report the quantity of alpha lost during implementation of the investment idea due to transaction costs. More recently, Bertsimas and Lo (1996), Almgren and Chriss (1999, 2000), Kissell, Glantz, and Malamut (2004) introduced a framework to minimize market impact and transaction costs, as well as a process to determine appropriate optimal execution strategies. These efforts have helped provide efficient implementation—the process known as algorithmic trading¹.

While empirical evidence has shown that when properly specified, algorithms result in lower transaction costs, the process necessitates investors be more proactive during implementation than they were previously utilizing manual execution. Algorithms must be able to manage price, size, and timing of the trades, while continuously reacting to market condition changes.

¹A review of market microstructure and transaction cost literature is provided in Chapter 2, Market Microstructure.